

Database Concepts, Importing Data & Mastering the SELECT Statement

Learn how data is organized in enterprise databases and retrieve it like a Data Engineer.

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Today's Agenda

- ✓ Quick Recap
- ✓ Database vs Schema vs Table
- ✓ Understanding Dot Notation (Database.Schema.Table)
- ✓ Creating an HR Schema
- ✓ Creating Tables using Schema
- ✓ Importing Data from CSV Files
- ✓ SELECT *
- ✓ Specific Columns
- ✓ Aliases (AS)
- ✓ CONCAT()
- ✓ DISTINCT
- ✓ TOP
- ✓ Assignment
- ✓ SQL Best Practices

Quick Recap

- ✓ Learned What a Table Is
- ✓ Created Our First Table
- ✓ Inserted Sample Data
- ✓ Verified the Data
- ✓ Retrieved Required Information Using SELECT

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What is a Database?

A database is a organized collection of related data that allows information to be stored, managed, and retrieved efficiently.

Imagine You Own a Company...

As your company grows, you'll store information such as:

-  Employees
-  Customers
-  Products
-  Orders
-  Sales
-  Projects

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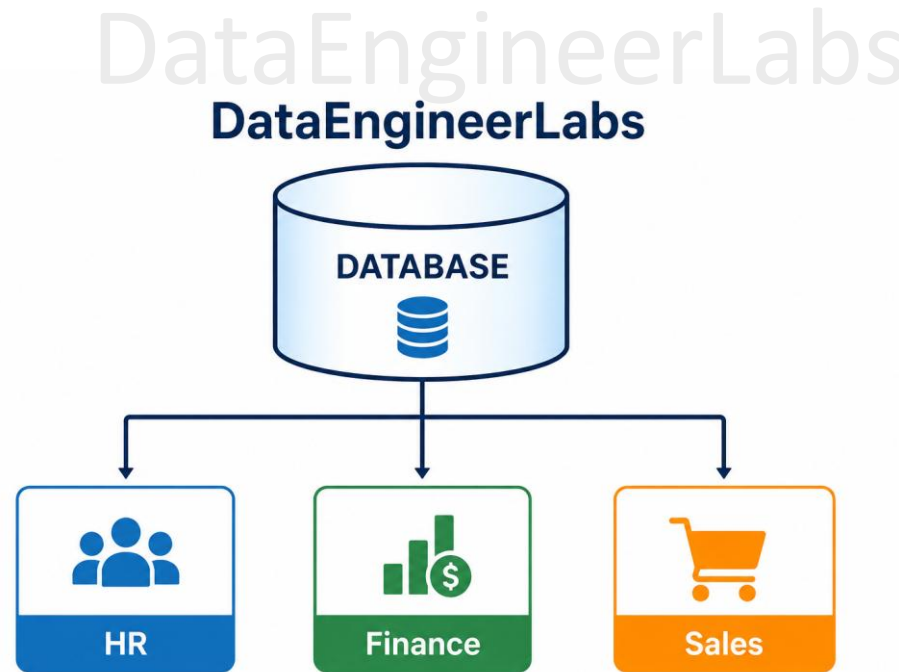
 Database
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	Employees
	Customers
	Products
	Orders
	Projects

What is a Schema?

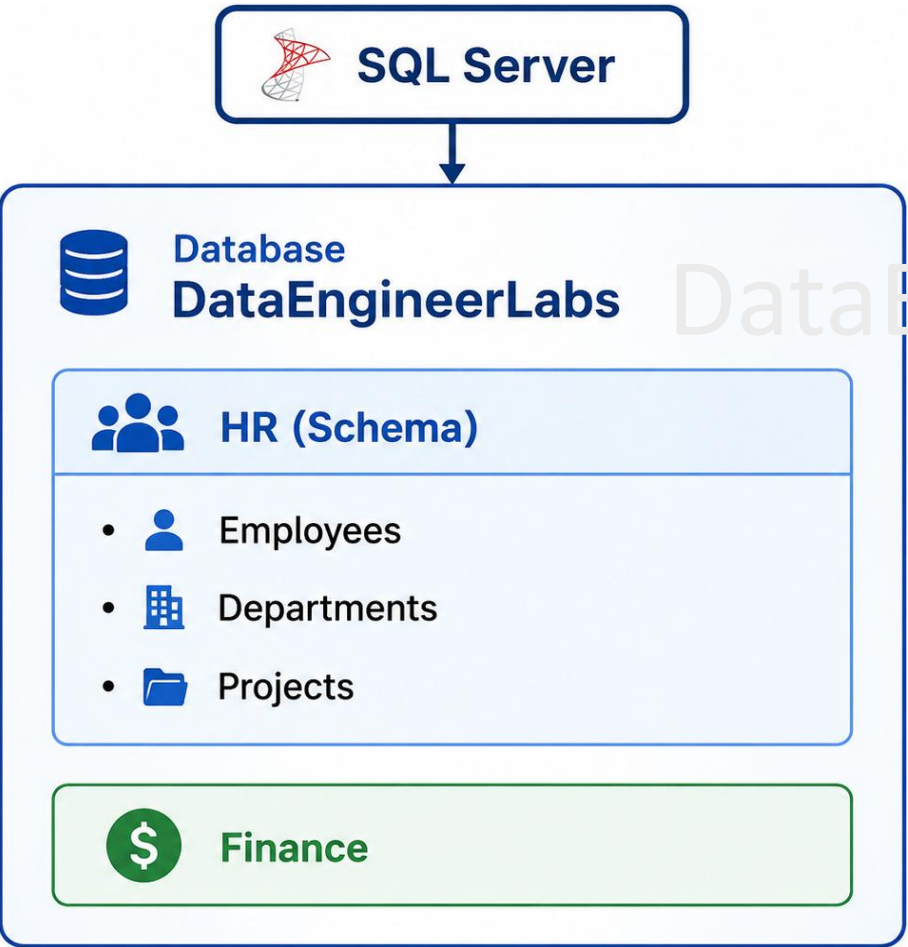
A Schema is a logical grouping of related database objects within a database. It helps organize, secure, and manage data more effectively.

Example: HR, Finance and Sales are common schemas in enterprise databases.



What is a Table?

A table is an organized collection of related information stored in rows and columns.



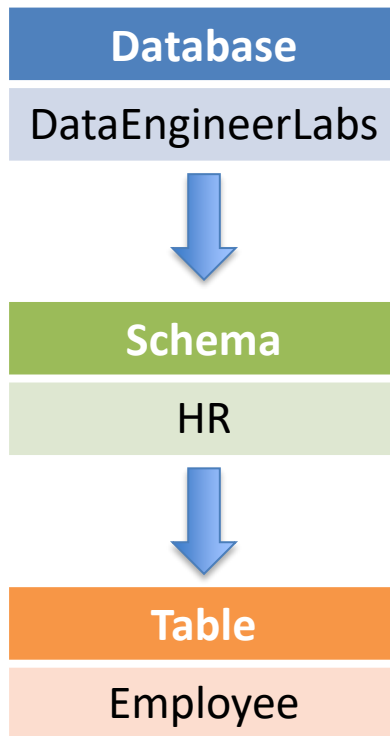
ID	Name	Department	Salary
1	John Smith	HR	60000
2	Emily Johnson	IT	75000
3	Michael Brown	Finance	70000
4	Sarah Davis	Marketing	65000
5	David Wilson	Sales	68000

Understanding Dot Notation

In SQL Server, every database object has an address, just like every house has a postal address. Instead of using a street, city, and country, SQL Server identifies objects using:

Database.Schema.Table

Each part of the name tells SQL Server exactly where to find the table.



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Create HR Schema and Tables

```
USE DataEngineerLabs;
```

```
GO
```

```
CREATE SCHEMA HR;
```

```
GO
```

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Create table and load data from CSV

Execute the table creation script

Create_Employees.sql

Import Employees.csv using BULK INSERT

```
BULK INSERT HR.Employees
FROM 'path\to\Employees.csv'
WITH (
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
```

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Mastering the SELECT Statement

1. Let's say HR wants to see every piece of information stored about employees
2. HR only needs the employee's First Name, Last Name, and Salary
3. The HR report should display more user-friendly column names
4. The HR Manager wants a Full Name column instead of separate First Name and Last Name columns
5. Management wants to know how many departments have/had employees in the company
6. The HR team wants to preview only the first few employee records.

Episode Summary

- ✓ Retrieved all columns using SELECT *
- ✓ Selected specific columns
- ✓ Renamed columns using AS
- ✓ Combined values using CONCAT()
- ✓ Removed duplicates using DISTINCT
- ✓ Limited results using TOP

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Assignment

Complete the following SQL tasks using the **HR** database.

Question 1

Display **all records** from the HR.Employees, HR.Projects and HR.Departments table.

Question 2

Display only the following columns:

EmployeeID

FirstName

LastName

DepartmentID

Question 3

Create a report showing:

Employee ID

Full Name (using CONCAT())

Email Address

Rename the Full Name column as **Employee Name**.

Question 4

Display a list of **unique Department IDs** available in the Employees table.

Question 5

Create following report using HR.Employee table :

Employee ID	Employee Name	Job Title	Salary
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Data Engineer Tip



Data Engineer Best Practice

- ✓ Avoid using SELECT * in production.
- ✓ Retrieve only the columns you need.
- ✓ Use meaningful aliases.
- ✓ Format SQL for better readability.
- ✓ Practice every query yourself.

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Thank You

Keep Building. Keep Experimenting

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